The Twilight of the Chatbots

Ethan Mollick — One Useful Thing • June 30, 2026

oneusefulthing.org/p/the-twilight-of-the-chatbots

> This document is an archival summary of Ethan Mollick's June 30, 2026 post on his Substack, "One Useful Thing." Content is reconstructed from the original post and contemporaneous reporting. Mollick is a Professor of Management at Wharton with a research focus on entrepreneurship and innovation, and is the author of *Co-Intelligence: Living and Working with AI* (2024).

Central Thesis

We are living through the twilight of the chatbot as the primary interface for AI — and with it, the end of the largest bottleneck between AI capability and AI utility. The argument is that AI models have for some time been capable of far more than their chatbot interfaces allowed most people to access. The shift to agent-based interaction is not primarily a story about better models; it is a story about better interfaces unlocking latent capability that was already there.

> *"We are moving from a world where non-experts use chatbots to fill in gaps to one in which experts use agents to get work done."*

The Capability Overhang Concept

Mollick's core reframe: the gap between what AI can do and what most users experience is not primarily a capability gap. It is an **interface gap**. AI capability has been running ahead of AI accessibility.

Capability Overhang Defined

Models have been smart enough to do extraordinary things for some time. The bottleneck has been the chatbot format itself — a synchronous, turn-by-turn interface that imposes cognitive costs on users, requires them to maintain context and continuity across interactions, and limits the time horizon of any single task. Most people's experience of AI

> is therefore significantly behind what the models can actually deliver, because they are accessing frontier intelligence through an interface designed for FAQ retrieval.

This reframing carries a significant practical implication: as interfaces improve, observed capability improvements will appear to accelerate — not because the underlying models are improving faster, but because users will finally be accessing capability that was latent all along. Each new interface paradigm will feel like a leap in capability because, in terms of accessible utility, it is.

Why Chatbots Cap Utility

The chatbot format has specific structural weaknesses that suppress AI usefulness, independent of model quality:

- **Synchronous constraint:** The user must remain present, actively directing each step. This limits task length to what a human can sustain in a single session and makes the human the pacing bottleneck.
- **Cognitive cost per turn:** Each exchange requires the user to formulate a prompt, evaluate a response, and decide next steps. The interface itself generates cognitive overhead that scales with task complexity, often overwhelming the time savings the AI provides.
- **No tool access or environment:** Chatbots interact with text. They cannot run code, browse file systems, inspect their own errors, re-run tests, or take actions in the world. The intelligence is real; the reach is not.
- **No persistent state:** Context resets between sessions. Long-horizon tasks that require memory across interactions are structurally unsupported.

These are interface failures, not model failures. A highly capable model accessed through a chatbot will underperform its potential the same way a powerful computer accessed through a teletype terminal underperforms its potential.

What Agents Change

Agents are characterized by three additions that directly address chatbot limitations:

Chatbot	Agent
Synchronous, human-paced	Asynchronous — works while human is absent
Text in, text out	Access to tools: file system, terminal, browser, APIs
Session-bound context	Persistent state, long-horizon tasks
Human manages continuity	Agent inspects its own failures, loops until done
Human specifies each step	Human specifies goal; agent finds the steps

> *"A model that can work for hours, inspect its own failures, run tests, read screenshots, and keep moving is closer to an asynchronous worker than a chatbot."*

Mollick specifically references Claude Fable 5 (subject to government access restrictions at time of writing) as demonstrating 9-hour autonomous work sessions on complex software projects that would have taken a team well over a week using traditional approaches. This represents a different category of AI use — not assistance with human work, but delegation of human work.

The Manager Stance

The practical framework Mollick proposes for working with agents is the **manager stance**: the human's job shifts from doing each step to specifying goals, reviewing outputs, and correcting the final result. This is the role of a manager relative to a capable employee — not micromanagement of process, but ownership of outcomes.

The Manager Stance in Practice

1. **Commissioning:**

   Provide a clear objective, relevant context, and quality standards — not a sequence of steps.
2. **Reviewing:**

   Inspect outcomes against the brief rather than supervising execution in real time.
3. **Correcting:**

Direct the agent to revise specific aspects of the result, not to redo individual steps.

4. **Validating:**

   Apply domain judgment to catch errors that the agent cannot self-detect — particularly in the "jagged" areas where model capability drops unexpectedly.

The cognitive load shifts from execution to evaluation. Domain expertise becomes more valuable, not less: it is the resource that makes validation meaningful.

Notably, this is also where the limits of purely autonomous operation become visible. Better models yield more output, but checking whether that output is correct doesn't get cheaper at the same rate — and is bounded by what the human actually knows in the domain. The manager stance requires genuine expertise to exercise properly.

Acceleration and Access Restrictions

Mollick notes that frontier model capabilities are accelerating. Better models from leading American labs have been releasing faster than at any prior point. He also flags a new structural constraint: government access restrictions have halted public access to two of the most capable current models — Claude Fable 5 and GPT-5.6 — at time of writing, creating a gap between frontier capability and what the broad market can access.

This introduces a new dimension to the capability overhang: it is now partly policy-mediated. The models exist; access is controlled. The practical implication is that organizations with regulated access to frontier models have a structural AI advantage that is neither market-acquired nor merit-based.

Interface as the Variable

The through-line of the piece is that interface design is the underappreciated variable in AI impact. The worldview Mollick developed in prior work — that AI's effect is concentrated in how well it can be accessed and applied by domain experts — extends naturally: if the interface is the bottleneck, then the most important near-term AI developments may be interface shifts (chatbot → agent, agent → ambient) rather than model capability improvements.

This has direct implications for organizations evaluating AI strategy: decisions made around chatbot-accessible capability may systematically underestimate what agentic interfaces make possible, and vice versa. A "no meaningful economy-wide effect yet" conclusion reached while the dominant interface is chatbots may reverse quickly as agent interfaces diffuse.

Connection to Prior Mollick Frameworks

This piece extends two prior arguments:

- **Jagged Frontier (2023 BCG study):** AI is brilliant at some complex tasks and poor at seemingly simple ones; the pattern is unpredictable without direct experimentation. The manager stance in agent work is partly a practical response to the jagged frontier — the human's validation role exists precisely because the capability profile is uneven.
- **Novice Benefit (Brynjolfsson customer support study):** AI disproportionately lifts novice performance. In agent contexts, Mollick suggests the ceiling shifts too — but the validation bottleneck means expert judgment remains irreplaceable for high-stakes outputs.

Why This Matters for Organizations

1. **Chatbot benchmarks understate agentic potential.** ROI analyses and productivity studies conducted in chatbot contexts are not predictive of agentic outcomes. Organizations waiting for clearer ROI signals may be waiting on the wrong evidence base.
2. **Interface investment is now strategic.** The choice of AI interface — which tools, which environment, how much autonomy — is not a UX decision. It determines how much of available capability reaches users.
3. **The workforce impact thesis needs recalibration.** If capability overhang is real and interface improvement unlocks it, impact on knowledge work may arrive faster and more broadly than adoption curves based on chatbot-era data suggest.
4. **Domain expertise is the scarce validation input.** As agents do more, the bottleneck shifts to humans who can verify correctness in complex domains. This reinforces the case for retaining and developing domain experts rather than treating AI efficiency as a headcount reduction lever.

Ethan Mollick is a Professor of Management at the Wharton School of the University of Pennsylvania, specializing in entrepreneurship and innovation. He is the author of *Co-Intelligence: Living and Working with AI* (Penguin, 2024) and writes the "One Useful Thing" newsletter. This document archives his June 30, 2026 post at oneusefulthing.org/p/the-twilight-of-the-chatbots.